

Lab 3: Hawaii's Precipitation Time Series Forecasting Using Recurrent Neural Networks

In this lab, you will build RNN models yourself to perform precipitation time series forecasting given the historical precipitation time series of Hawaii. The data we will be using is from National Centers for Environmental Information (*Data is provided in an Excel sheet*).

Task 1: Precipitation Time Series Exploration and Visualization (20 pts).

Download the time series data, and save it in your Google Drive so that it can be accessed by Colab. Read the csv file with Pandas and visualize the time series from the data in which X axis is the time and Y axis is the monthly precipitation.

Hints: You need to convert the "Date" column into datetime with

```
import datetime

string = "189501"
element = datetime.datetime.strptime(string, "%Y%m")
```

Task 2: Convert this time series into a prediction dataset and perform train/test split (20 pts).

Convert this time series into a prediction dataset with input X (num_sample, num_len) and output Y (num_sample, num_len). Here, each sample in X indicates a num_len length time series and its corresponding label in Y indicates a num_len length time series with 1 month lag, e.g., X -> (P₁, P₂, ..., P₁₂) and Y -> (P₂, P₃, ..., P₁₃). Split this dataset into training and testing with 80:20 ratio and normalize the time series after doing the train/test split.

Task 3: Develop a TimeSeriesNN model class (20 pts).

Create a TimeSeriesNN model class which consists of a 3-layer LSTM layer (torch.nn.LSTM) with hidden size as 40 and one linear layer.

Task 4: Train a TimeSeriesNN model on the time series data (20 pts).

Train a TimeSeriesNN model on the created time series dataset by 100 epochs and show the train and test MAE for each epoch.

Hint:

- $MAE = \frac{\sum_{i=1}^n |P_o - P_p|}{n}$
- Create the train and test dataloader.
- The evaluation metric is *MAE* but not *RMSE*!

Task 5: Visualize the predicted time series together with the ground truth (20 pts).

Visualize the predicted time series together with the ground truth to show the performance difference.

To submit:

- Put your Google Colab link in GitHub classroom and then submit